

Gilbert Chang

U.S. Citizen | ITAR Eligible | Rising Senior Summer 2026 | +1 (925)-444-5802 | chang940@purdue.edu | gilcha.com

EDUCATION

Purdue University

August 2023 – May 2027

Bachelors of Science in Mechanical Engineering and Computer Science

GPA: 3.70/4.00

Awards: Kenneth Ralph Scudder Mechanical Engineering Scholarship

Relevant Coursework: Heat and Mass Transfer, Measurement and Control Systems II, Machine Design, Mechanics of Materials, Signals and Systems, Fluid Mechanics, Thermodynamics, Electrical Engineering Fundamentals II.

EXPERIENCE

Persona AI

May 2025 – August 2025

Mechanical Engineering Intern

Houston, TX

- Re-engineered NASA Robonaut 2 hand: remodeled 50+ CAD parts (Onshape) and cut joint width 11%; released ASME Y14.5 drawings & URDF, presented design reviews, and delivered an ahead-of-schedule MJF prototype; human-factor requirements (grasp/force targets) informed by MIL-STD-1472H.
- Designed a cable-tension sensor for servo/ball-screw control using closed-form strain/signal models with FEA correlation; optimized geometry to meet 3x FoS; released drawings, managed sourcing, and visited vendors.
- Developed and commissioned a modular joint-angle test fixture; integrated encoders & signal conditioning; authored test procedures, evaluated drift under stray MIG magnetic fields, documented fixes.
- Authored a YAML-driven Python/JAX kinematics suite that computes 100k FK/IK poses in ≤ 0.1 s, auto-optimizes encoder gaps, and estimates sensor stresses; reduced study time by 90%.

CoMMA Lab, Purdue University

November 2024 – Present

Undergraduate Research Assistant

West Lafayette, IN

- Leading research effort in collaboration with UT Austin MERGe Lab to develop FEA-based (Abaqus) optimizer to route air channels for fluidic innervation in soft robotic grippers to optimize slip/force algorithm performance.
- Engineered polyurethane soft robotic gripper and test fixture; embedded strain gauges & flex sensors; identified root-cause of sensor noise during operation, re-engineered Wheatstone bridges and low-pass filters to attenuate robot vibration; optimized gripper FDM printed mold designs and post-processing for silicone casting release.
- Authored mesh processing pipeline (Python/C++) that converts URDF meshes to sphere-trees, accelerating robot simulation runtimes by 10%; conference paper submitted to IEEE IROS 2025 (advised by Zachary Kingston).

Cai Research Group, Herrick Laboratories

January 2025 – May 2025

Undergraduate Research Assistant

West Lafayette, IN

- Developed Python/MATLAB thermal system-identification pipeline for a psychrometric test chamber (room dynamics, thermocouples, heat pump); leveraged the CVX optimization toolbox to fit open-loop test data to lumped-sum RC network transient thermal models, generating models for formal analysis and controller design.
- Validated thermal models in closed-loop chamber experiments: simulated temperature trajectories held within $\pm 5^\circ\text{C}$ of measured values across the full operating range, confirming high-fidelity replication of chamber dynamics; maintained validation schedule, reported progress to principal investigator, graduate students, and collaborators.

PROJECTS

Baja SAE – Composites Subteam | (*Materials Testing & Manufacturing*)

- Fabricated carbon fiber body panels, fairings, steering wheel via forging and VARTM; assisted with car coast-down testing, identifying root causes of drag differences between Ansys Fluent simulations and real world performance.
- Integrated composite assemblies with chassis under high-cycle fatigue constraints; evaluated various fasteners to enable a complete body panel swap in 60 seconds; rapidly iterated on nosecone topology to slash drag by 30%.
- Developed forged-carbon dogbone, mold (Aluminum, PLA), and UTM fixture (SolidWorks); captured CSV strain data from Instron UTM and fit modulus curve in MATLAB for Ansys forged-carbon material parameters.

TECHNICAL SKILLS

CAD & Drawings: Siemens NX, SolidWorks, Onshape; drawing release; Aras Innovator; KiCad

Programming & Data: Python (JAX, NumPy, Pandas, PyYAML, PySerial), C/C++, MATLAB; Git, Linux

Prototyping & Test: CNC mill/lathe, sheet metal, 3D printing (FDM/SLS), VARTM/forged composites; load cells, strain gauges (metal foil, semiconductor)/bridges, power supplies, oscilloscopes, multimeters, UTM

Analysis & Simulation: FEA (Abaqus, Ansys), CFD (Ansys Fluent); tolerance stackups; first-principles modeling

Standards & Methods: GD&T (ASME Y14.5), DFM/DFA, Limits & Fits, MIL-STD-1472H